



Dr. M. S. Sheshgiri Campus, Belagavi

**Department of
Electronics and Communication Engineering**

**OPEN ENDED Report
on
ADAPTIVE CRUISE CONTROL**

By:

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Under the Guidance of

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that project entitled “**Adaptive Cruise Control**” is a bonafide work carried out by the student team of “**Chandrashekhar Bhavi-02FE22BEC020, Jamuna Pulpatli-02FE22BEC032, Priyanka Budihal-02FE22BEC054, Pruthviraj Banoshi-02FE22BEC058**”. The project report has been approved as it satisfies the requirements with respect to the minor project work prescribed by the university curriculum for B.E. (VI Semester) in Department of Electronics and Communication Engineering of KLE Technological University Dr. M. S. Sheshgiri CET Belagavi campus for the academic year 2025-2026.

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ABSTRACT

This project implements a basic Adaptive Cruise Control (ACC) system using an Arduino Mega, designed to maintain a safe following distance from objects ahead. The system utilizes an HC-SR04 ultrasonic sensor to measure the distance to a leading object and adjusts the speed of two DC motors accordingly through an L298N motor driver. A potentiometer allows the user to manually set a base speed, while the system dynamically reduces speed or stops the motors based on proximity to obstacles. When no object is detected, the vehicle moves at the set base speed. The project demonstrates a simple yet effective way to simulate intelligent speed control for autonomous or semi-autonomous robotic vehicles.

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Chapter 1

Introduction

1.1 Problem Statement

Adaptive Cruise Control (ACC) is an advanced driver-assistance system that automatically maintains a safe distance between a vehicle and the one ahead. Traditional cruise control systems cannot adapt to changing traffic conditions, requiring constant driver intervention. The problem is to develop an ACC system that uses sensors like radar or lidar to monitor the distance and relative speed of the lead vehicle. Based on this data, the system should adjust the ego vehicle's speed through acceleration or braking. The goal is to improve driving safety, reduce driver fatigue, and ensure smooth vehicle operation in dynamic traffic environments.

1.2 Introduction

Adaptive Cruise Control (ACC) is an advanced driver assistance system designed to automatically adjust a vehicle's speed to maintain a safe distance from the vehicle ahead. Unlike traditional cruise control systems that keep a constant speed, ACC uses sensors such as radar, lidar, and cameras to monitor surrounding traffic conditions in real time. When it detects a slower-moving vehicle in the same lane, it reduces the vehicle's speed by decelerating or applying the brakes. Once the path is clear, it accelerates back to the preset cruising speed. This functionality enhances safety, reduces driver fatigue during long drives or stop-and-go traffic, and provides a smoother driving experience. ACC is increasingly being integrated into modern vehicles and plays a vital role in the development of semi-autonomous and autonomous driving technologies.

1.3 Objectives

- Maintain a safe distance from the vehicle ahead.
- Reduce driver fatigue in long or congested trips.
- Automatically adjust speed based on traffic flow.
- Enhance driving comfort and convenience.
- Improve road safety by preventing rear-end collisions.
- Optimize speed control for better fuel efficiency.
- Support semi-autonomous vehicle functions.
- Integrate seamlessly with other driver assistance systems.

1.4 Applications

- Autonomous Robots: Helps mobile robots maintain a safe distance from obstacles or other robots while moving.
- Self-Driving Vehicles (Basic Concept): Demonstrates foundational principles of adaptive speed control based on sensor input.
- Obstacle Avoidance Systems: Used in robots or vehicles to prevent collisions by adjusting speed or stopping.
- Educational Projects: Ideal for teaching concepts of sensors, motor control, and feedback systems in embedded systems courses.
- Automated Guided Vehicles (AGVs): Maintains safe following distance in warehouse or factory automation.
- Assistive Driving Aids: Forms a basis for low-cost proximity alert systems in vehicles or robotics.

Chapter 2

Design Specification

2.1 Simulation

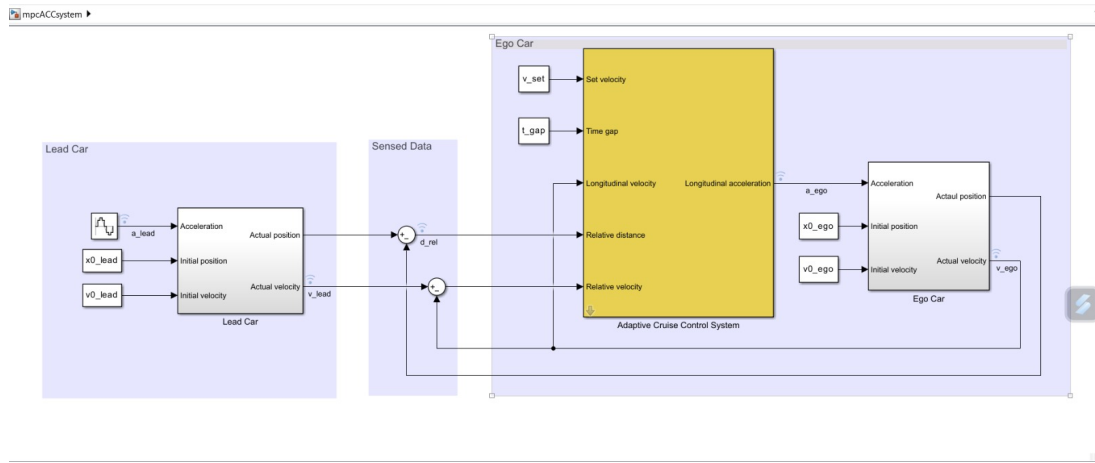


Figure 2.1: Simulation

The Simulink model simulates an Adaptive Cruise Control (ACC) system using a Model Predictive Control (MPC) strategy. It includes two vehicles: a Lead Car, which receives acceleration, initial position, and velocity as inputs to generate its motion; and an Ego Car, which is controlled by the ACC system. The model computes relative distance and relative velocity between the two cars using sensed data, which, along with the ego car's speed, desired set speed, and a predefined time gap, is sent to the ACC controller. The controller calculates the required acceleration to maintain a safe following distance. This setup effectively demonstrates how an ACC system responds to traffic dynamics, helping validate control strategies in autonomous driving applications.

2.2 Hardware

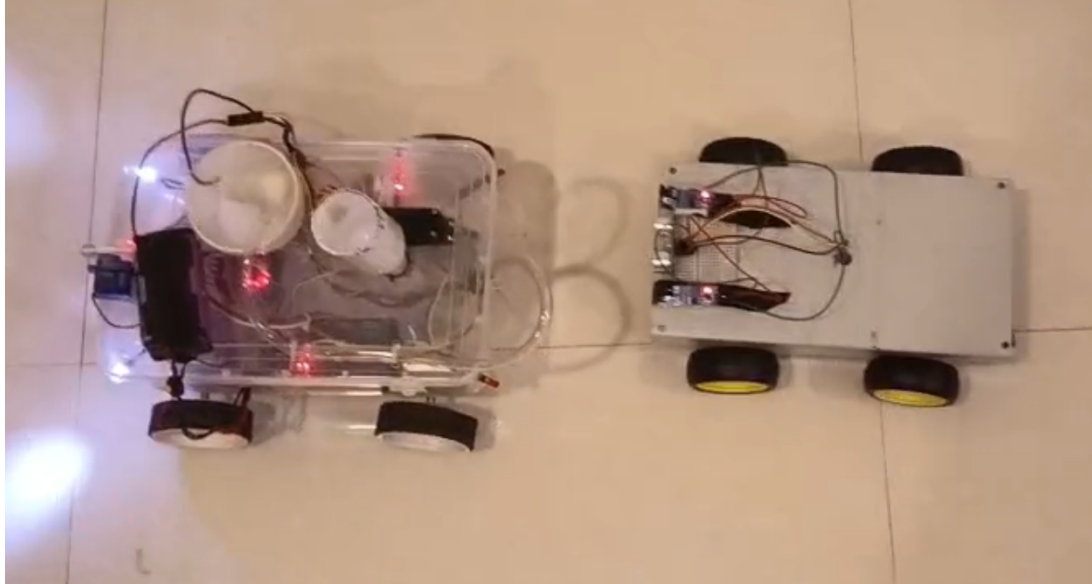


Figure 2.2: Hardware model of Adaptive Cruise Control

2.3 Code

```
define ULTRASONIC_SENSOR_TRIG 11
define ULTRASONIC_SENSOR_ECHO 12

    int enableRightMotor = 8;
int rightMotorPin1 = 2;
int rightMotorPin2 = 3;

    int enableLeftMotor = 9;
int leftMotorPin1 = 4;
int leftMotorPin2 = 5;

    int potPin = A0;

    long duration;
int distance;
int baseSpeed;
int followSpeed;
    void setup()
Serial.begin(9600);

    pinMode(enableRightMotor, OUTPUT);
pinMode(rightMotorPin1, OUTPUT);
pinMode(rightMotorPin2, OUTPUT);

    pinMode(enableLeftMotor, OUTPUT);
pinMode(leftMotorPin1, OUTPUT);
pinMode(leftMotorPin2, OUTPUT);

    pinMode(ULTRASONIC_SENSOR_TRIG, OUTPUT);
pinMode(ULTRASONIC_SENSOR_ECHO, INPUT);

    stopMotors();
}

    void loop() {
int potValue = analogRead(potPin);
baseSpeed = map(potValue, 0, 1023, 0, 255);
distance = getDistance();

    Serial.print("Distance: ");
Serial.print(distance);
Serial.print(" cm — Base Speed: ");
Serial.println(baseSpeed);

    if (distance < 10 || distance >= 50)
float factor = (50 - distance) / 40.0;
```

```

followSpeed = baseSpeed * factor;
moveForward(followSpeed);
}
else if (distance < 5 distance >= 10)
followSpeed = baseSpeed * 0.3;
moveForward(followSpeed);
}
else if (distance < 0 distance >= 5)
stopMotors();

else
moveForward(baseSpeed);
}

    delay(100);
}

    int getDistance()
digitalWrite(ULTRASONICSSENSORTRIG, LOW);
delayMicroseconds(2);
digitalWrite(ULTRASONICSSENSORTRIG, HIGH);
delayMicroseconds(10);
digitalWrite(ULTRASONICSSENSORTRIG, LOW);
duration = pulseIn(ULTRASONICSSENSORECHO, HIGH, 20000);
returnduration * 0.034/2;
}

    void moveForward(int speed)
analogWrite(enableRightMotor, speed);
analogWrite(enableLeftMotor, speed);

    digitalWrite(rightMotorPin1, HIGH);
digitalWrite(rightMotorPin2, LOW);

    digitalWrite(leftMotorPin1, LOW);
digitalWrite(leftMotorPin2, HIGH);
}

    void stopMotors()
analogWrite(enableRightMotor, 0);
analogWrite(enableLeftMotor, 0);

    digitalWrite(rightMotorPin1, LOW);
digitalWrite(rightMotorPin2, LOW);
digitalWrite(leftMotorPin1, LOW);
digitalWrite(leftMotorPin2, LOW);
}

```

Chapter 3

Result and Conclusion

3.1 Result

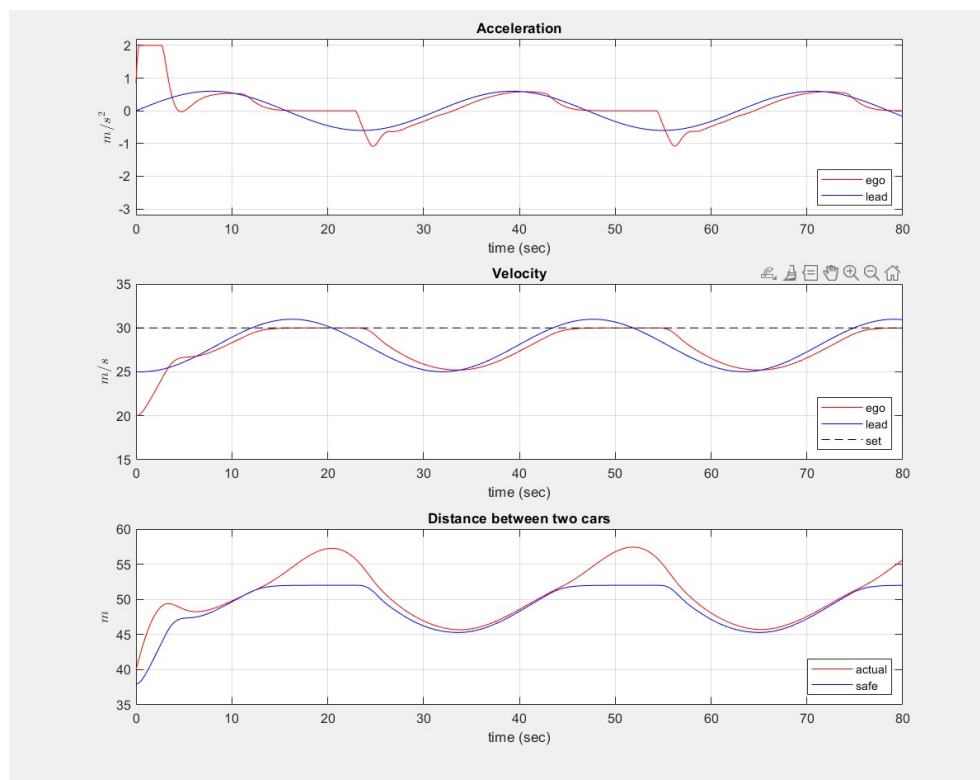


Figure 3.1: Output

3.2 Conclusion

Adaptive Cruise Control (ACC) is a vital advancement in modern vehicle technology that significantly enhances driving safety, comfort, and convenience. By intelligently adjusting vehicle speed to maintain a safe distance from other vehicles, ACC reduces driver workload and contributes to smoother traffic flow. As a key component of advanced driver assistance systems, ACC not only improves the overall driving experience but also lays the groundwork for future developments in autonomous driving.